Interpretable Head-Pose Features for Cervical Dystonia Screening from a Pre-Trained Body-Landmark Model

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Abstract

Cervical dystonia is a painful neurological movement disorder in which neck muscles contract involuntarily and hold the head in an abnormal posture: turned, tilted, or pushed forward or back. Its severity is usually scored by a clinician from visual observation. In this project we ask whether a pre-trained *body-landmark model, with no training of our own, provides enough geometry to flag such postures from an ordinary photograph.*

Using five landmarks from MediaPipe PoseLandmarker (nose, ears and shoulders), we design four scale-invariant features: head-tilt angle, nose midline offset, ear height asymmetry and shoulder inclination. A transparent rule table maps them to normal*,* rotational torticollis *or* laterocollis*. On four test photographs the nose offset separates the turned head from normal poses by a factor of 7, and ear asymmetry and shoulder inclination separate the tilted head by factors of 28 and 9. The rule classifies all four images correctly; because its thresholds were set on these same images, we treat this as a proof of concept and outline what a proper evaluation requires.*

1. Introduction

Clinical context. Dystonia is characterised by sustained or intermittent muscle contractions that cause abnormal postures [1]. In the neck, the main patterns are named by the direction of head deviation (Fig. 1): forward flexion (anterocollis), backward extension (retrocollis), rotation about the vertical axis (rotational torticollis) and lateral tilt toward a shoulder (laterocollis). Objective, camera-based measurements could support follow-up between clinic visits.

Our approach. We use the pre-trained MediaPipe PoseLandmarker [2, 4] as a fixed measurement device and design features from its landmarks. Each feature corresponds to a clinically meaningful deviation, so its value can be read and checked by a person, which matters more than raw accuracy in a screening setting. We target the two patterns visible in a frontal photograph: rotation (yaw) and lateral tilt (roll).

2. Related Work

Pose estimation. OpenPose [3] popularised real-time multi-person 2-D keypoints. BlazePose [2], which underlies MediaPipe Pose, predicts 33 body landmarks, including face points, on mobile hardware.

Feature engineering on keypoints. Normalised distances and angles between keypoints are a common way to turn pose estimates into interpretable measurements. Normalising by a body length, here the shoulder width, removes the dependence on image scale and camera distance.

3. Method

3.1. Landmarks

PoseLandmarker (Heavy) runs in static-image mode with detection, presence and tracking confidence 0.4. Of its 33 landmarks we use five: nose (0), left ear (7), right ear (8), left shoulder (11) and right shoulder (12). They are converted to pixel coordinates, with y increasing downward, and all distances are divided by the shoulder width $w = \|\mathrm{LS} - \mathrm{RS}\|$.

3.2. Features

F1 — head-tilt angle θ. The angle between $v = \mathrm{LE} - \mathrm{LS}$ and the upward vertical:

$$\theta = \arccos\left(\frac{v \cdot [0, -1]^\top}{\|v\|}\right). \tag{1}$$

In an upright posture the shoulder-to-ear vector points roughly upward, and lateral tilt rotates it. In a frontal photograph, however, the ear lies beside the head rather than above the shoulder, so $\theta \approx 27$–31° even for normal subjects. We therefore report F1 but do not use it in the decision.

F2 — nose midline offset Δx. Facing the camera, the nose sits above the shoulder midpoint m. Turning the head moves it sideways:

$$\Delta x = (\mathrm{N}_x - m_x)/w. \tag{2}$$

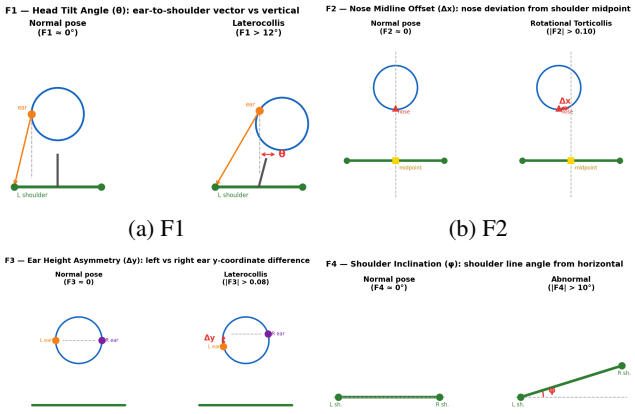
F3 — ear height asymmetry Δy. Both ears are level in a normal pose; laterocollis drops one ear toward its shoulder:

$$\Delta y = (\mathrm{LE}_y - \mathrm{RE}_y)/w, \tag{3}$$

where $\Delta y > 0$ means the left ear is lower.



Figure 1. Cervical-dystonia head postures. This work targets the two visible in frontal view: rotational torticollis (yaw) and laterocollis (roll).



(a) F1 (b) F2

(c) F3 (d) F4

Figure 2. Feature geometry. In each panel the left sketch is a normal pose and the right sketch an abnormal one.

Table 1. Decision rule, evaluated top to bottom. F1 is not used.

Prediction	Condition
Normal	$\lvert F2\rvert < 0.10$, $\lvert F3\rvert < 0.08$, $\lvert F4\rvert < 10^\circ$
Laterocollis	$\lvert F3\rvert > 0.08$ and $\lvert F4\rvert > 10^\circ$
Rot. torticollis	$\lvert F2\rvert > 0.10$ and $\lvert F3\rvert < 0.08$
Mixed / uncertain	otherwise

F4 — shoulder inclination φ. Laterocollis is often accompanied by elevation of one shoulder:

$$\varphi = \operatorname{atan2}(-(\mathrm{LS}_y - \mathrm{RS}_y),\ \mathrm{LS}_x - \mathrm{RS}_x)\,. \qquad (4)$$

F4 does not use head landmarks, so it gives a second, partly independent signal for lateral tilt.

3.3. Decision rule

Rotation shows up in F2; lateral tilt shows up in F3 and F4 together. Lateral tilt also shifts the nose, so laterocollis is tested *before* rotation (Table 1).

4. Results and Discussion

4.1. Test images

We evaluate on four frontal photographs: two normal upright poses, and the rotational-torticollis and laterocollis examples from Fig. 1.

4.2. Feature values

Table 2 lists the extracted features and Fig. 3 plots them against the thresholds.

F2 isolates rotation. Both normals have $|F2| \leq 0.051$, against 0.359 for the turned head, a $7\times$ margin.

F3 and F4 isolate lateral tilt. For laterocollis, $F3 = 0.279$ against ≤ 0.010 for normals ($28\times$), and $|F4| = 17.7^\circ$ against $\leq 1.9^\circ$ ($9\times$). Requiring both makes the decision robust to one noisy landmark.

Why F1 is excluded. The initial rule required $|F1| < 12^\circ$ for a normal pose, but frontal photographs give 27–31°, so both normals fell through to "mixed". It also tested laterocollis with $F1 > 12^\circ$, while the tilted head has $F1 = 3.4^\circ$, so that image was caught by the rotation test instead. Dropping F1 and testing F3 + F4 before F2 resolves both errors.

4.3. Qualitative results

Figure 4 overlays the landmarks and feature lines on each image.

5. Conclusion & Future Work

Four geometric features from a pre-trained pose model separate normal, rotated and laterally tilted head postures in our test images by large margins, and each feature can be read off the image by a person.

Limitations. The thresholds were set on the same four images used for evaluation, so there is no held-out accuracy and the results are a proof of concept. Pitch postures (antero- and retrocollis) are not covered by frontal 2-D features. The features assume a roughly frontal, level camera; camera roll shifts F3 and F4 together.

Table 2. Extracted feature values and decisions. Bold marks values beyond their threshold. "Initial rule" is a baseline that also required $|F1| < 12^\circ$ for *normal*.

Image	True pose	F1 tilt (°)	F2 nose offset	F3 ear asym.	F4 shoulder (°)	Our rule	Initial rule
Normal 1	Normal	26.67	−0.008	0.010	1.35	Normal ✓	Mixed ×
Normal 2	Normal	30.93	−0.051	0.007	1.92	Normal ✓	Mixed ×
Rotational	Rot. torticollis	37.71	**−0.359**	0.020	−0.78	Rot. torticollis ✓	Rot. torticollis ✓
Lateral tilt	Laterocollis	3.44	0.308	**0.279**	**−17.73**	Laterocollis ✓	Rot. torticollis ×

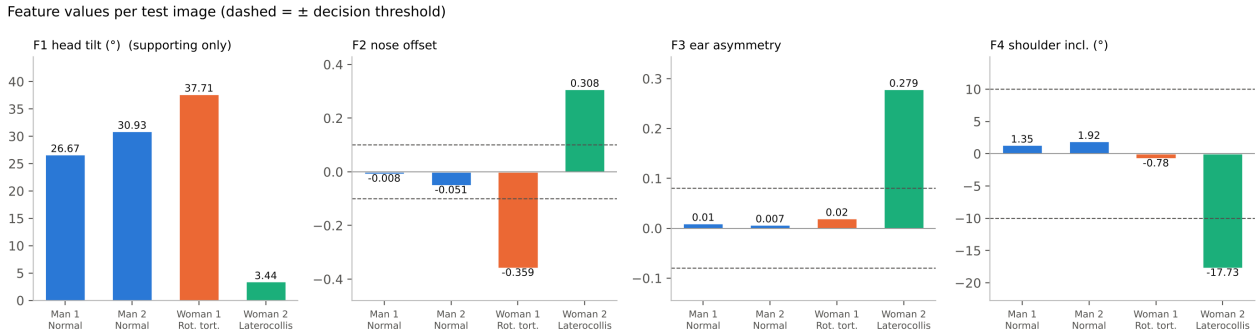
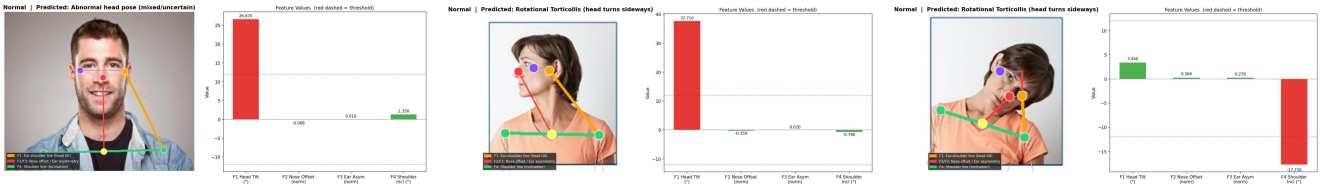


Figure 3. Feature values per test image. Dashed lines mark the ± decision thresholds; F1 has no threshold because it is supporting only.



(a) Normal 1 (b) Rotational torticollis (c) Laterocollis

Figure 4. Landmark overlays with feature bars (orange: ear–shoulder line; red: nose offset and ear asymmetry; green: shoulder line). Panel titles show the initial rule's output.

Next steps. (i) Collect labelled patient images or video frames and fit thresholds on one subset while testing on another; (ii) use MediaPipe's 3-D world landmarks to measure yaw, roll *and* pitch in degrees; (iii) normalise for camera roll with the hip line; and (iv) track features over video to quantify tremor and sustained deviation.

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References

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